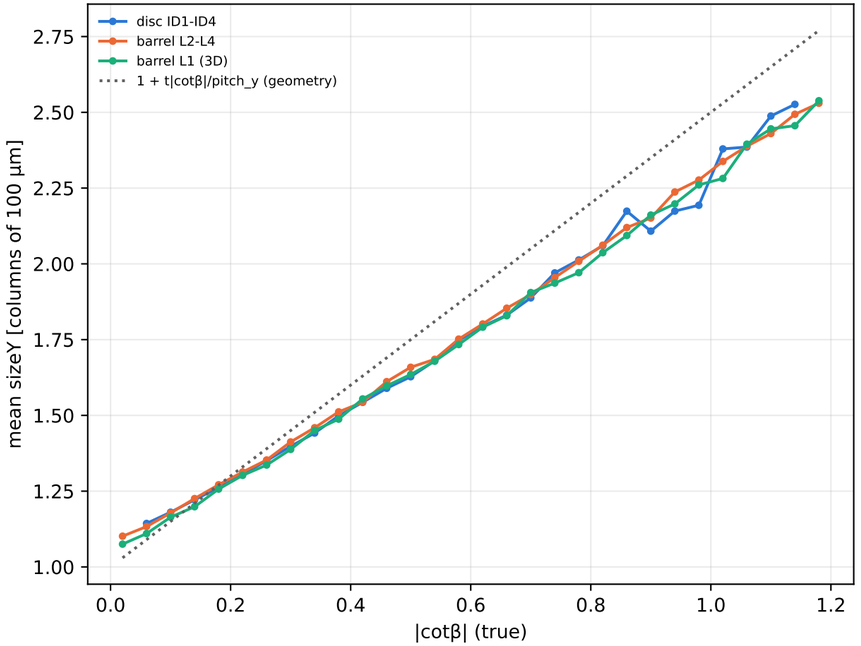
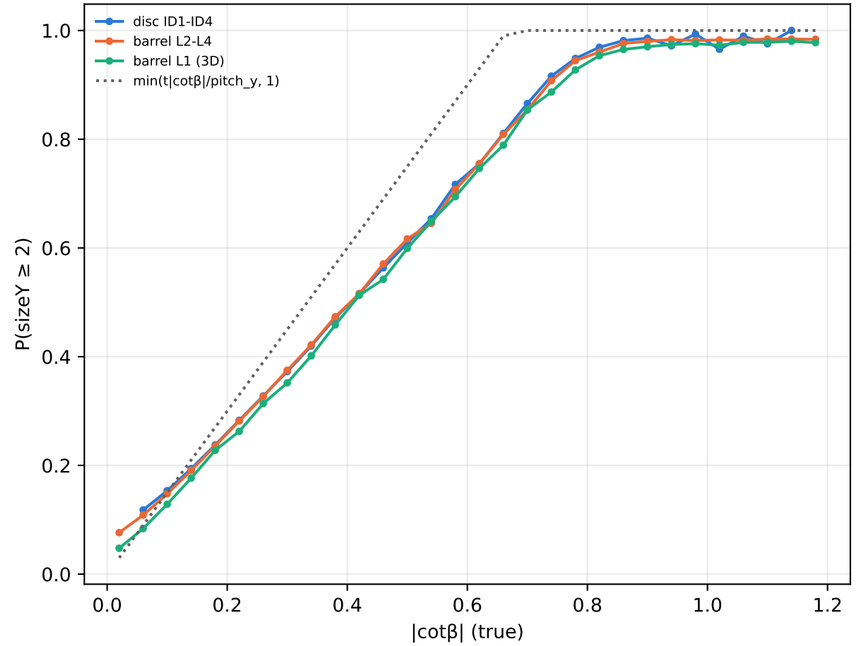
Page 1. Frames and cluster size vs incidence (consistent-linked, pT ≥ 1 GeV, 200 events; truth + clusters only)

```
LOCAL FRAMES (all 4000 modules; module axes in global coordinates)
every module: pitch x 25 um, pitch y 100 um, thickness 150 um;
  charge collected at w = -t/2 (Phase-2 digitizer, dir_z = -1)
barrel: y = -z on every module; w = +r or -r (zOutward), half each;
  x = +-phi; drift along +x, tanLA 0.202 (L2-L4), 0 (L1 3D)
disc ID1-ID4: y = -r (radially inward) on every module;
  w = +z or -z (facing), half each per ring; x = +-phi; no drift
=> beta = the plane (100 um pitch, normal) = the r-z plane on BOTH
prompt cotBeta = -zOutward z/r (barrel), -facing R/|Z| (disc)
a flip = a rigid 180 deg rotation (x, w reversed, y kept):
  the image <-> local-angle relation is identical for both facings;
  only WHICH local angles occur changes (disc: sign(beta) = -facing)
```

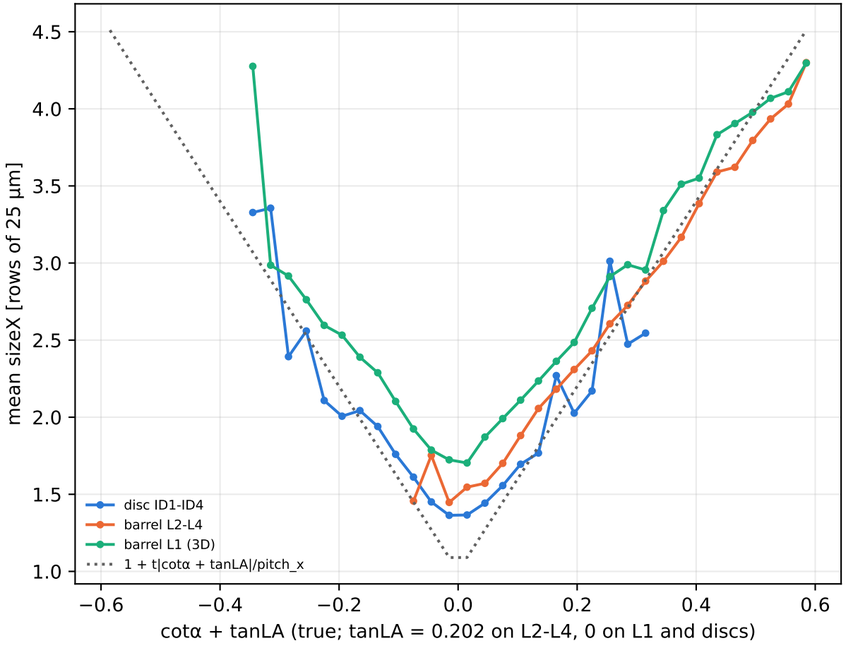
cluster length along local y vs |cotβ|



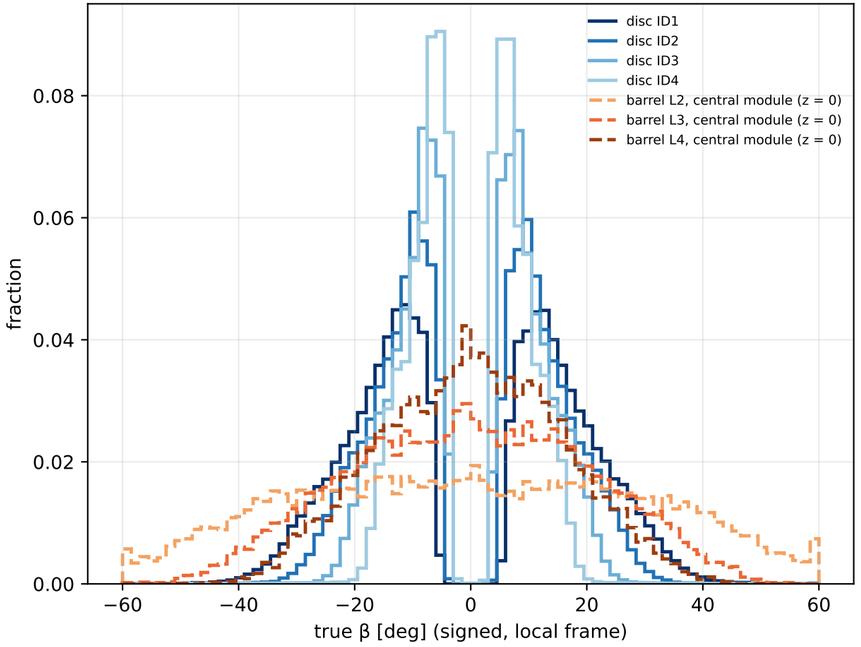
fraction of clusters spanning ≥ 2 columns



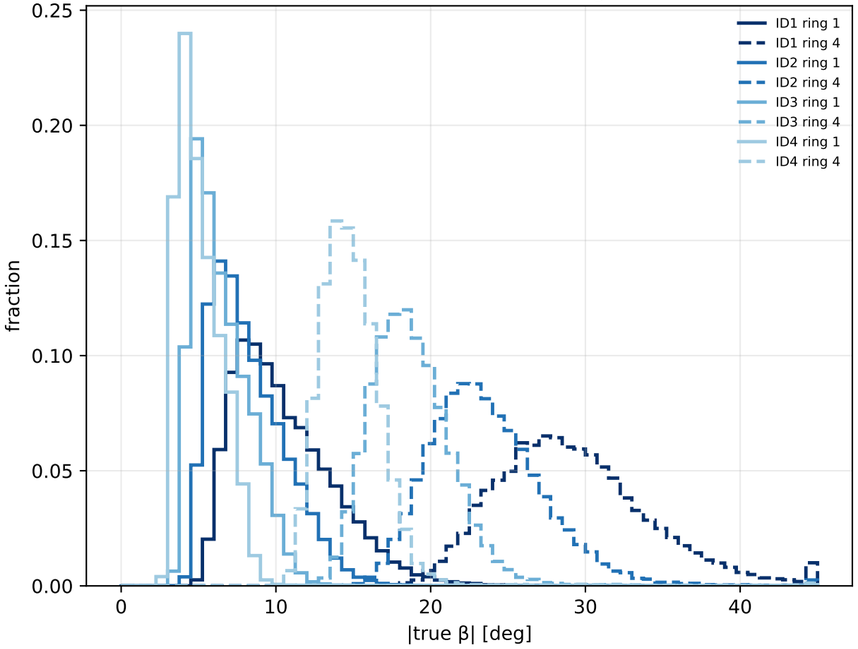
cluster width along local x: the drift only shifts x



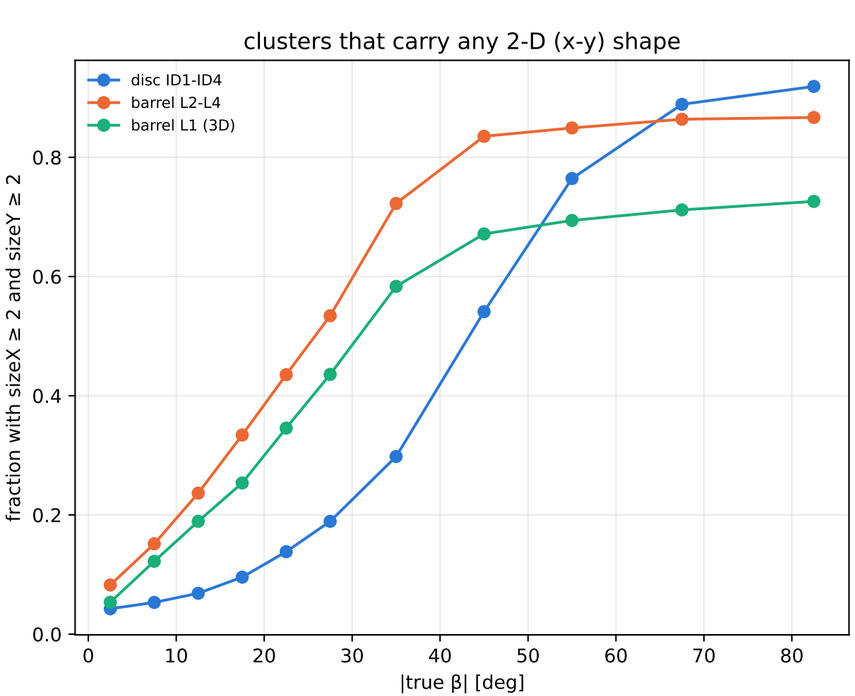
incidence: disc rings sit at ±R/|Z| (two facings);
central barrel modules are broad (vertex z spread / r)

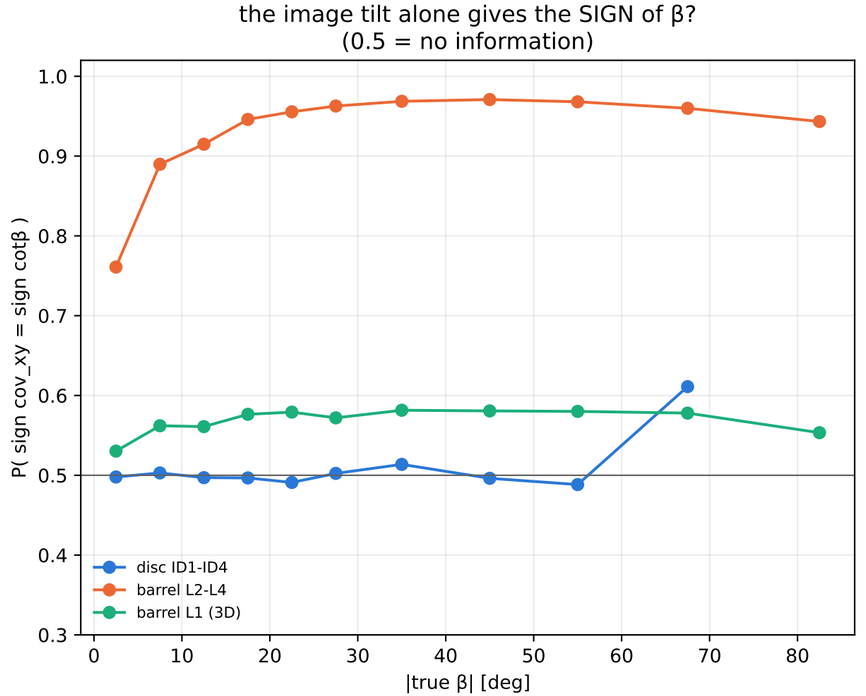


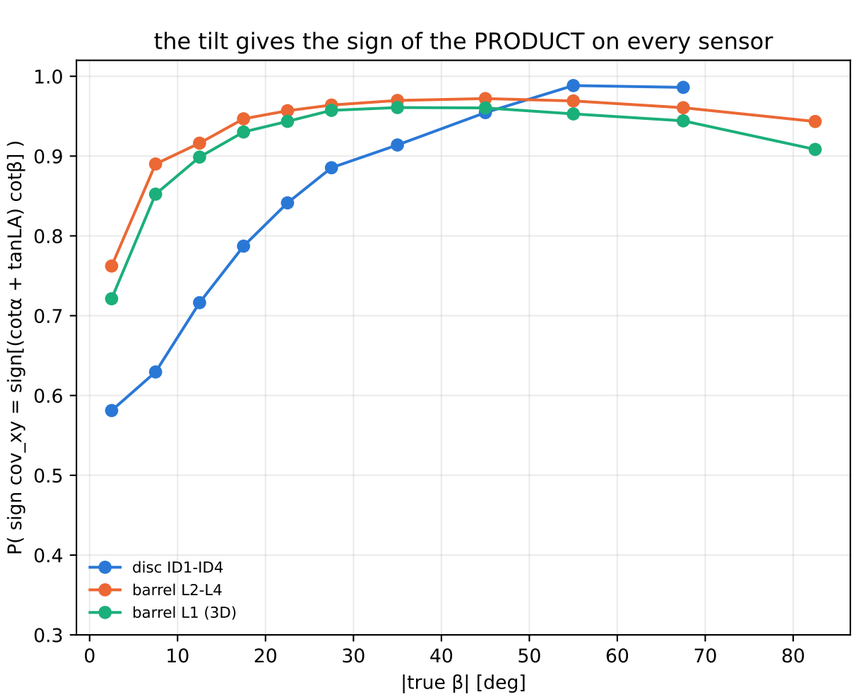
disc |β| per ring (ring 1 R = 5.3 cm, ring 4 R = 13.8 cm)



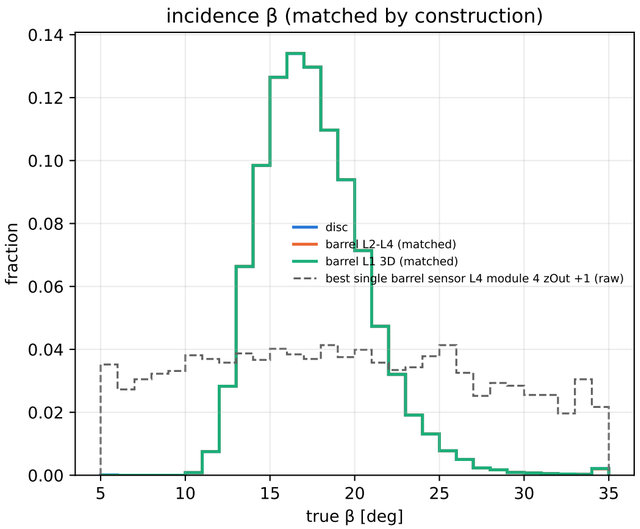
Page 2. A straight track deposits a segment along (cotα + tanLA, cotβ) in the (x, y) image. Without drift the image is symmetric under (α, β) -> (-α, -β): only sign(αβ) is visible.
With drift (L2-L4) cotα + tanLA > 0 for most tracks, so the tilt fixes sign(β). Charge-weighted covariance of the 24x16 ADC image, clusters with sizeX ≥ 2 and sizeY ≥ 2.

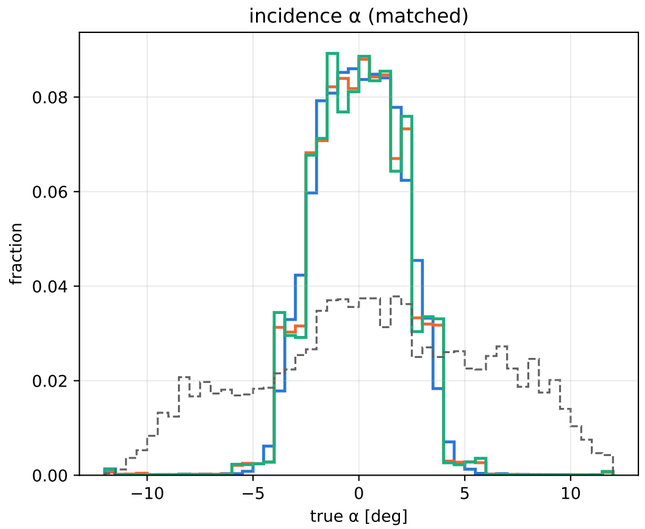


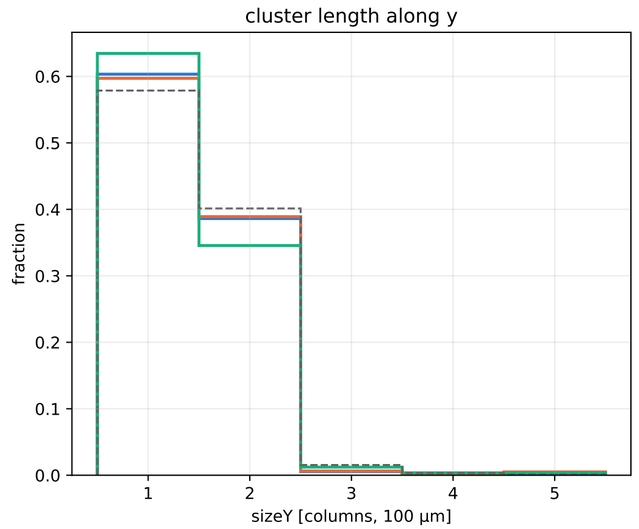


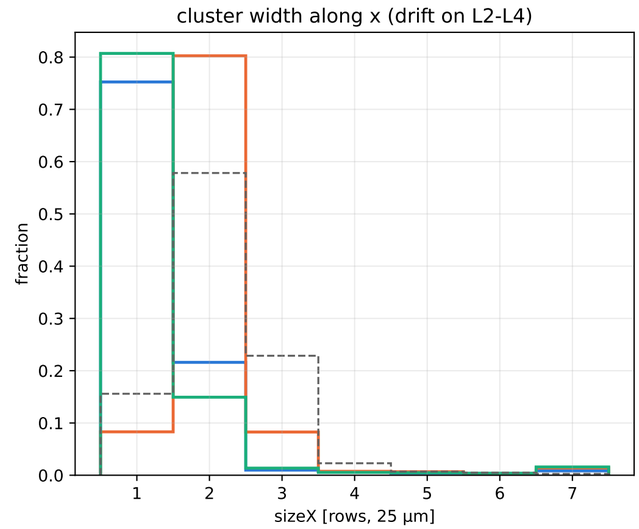


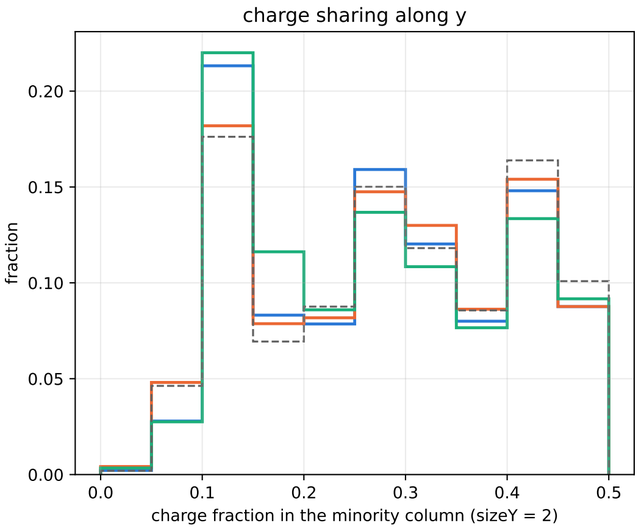
Page 3. Representative disc sensor ID2 ring 3 (R = 10.5 cm, |Z| = 32.9 cm), facing − (w toward the IP ⇒ β > 0) vs barrel sensors at MATCHED incidence (reweighted in true β × α). consistent-linked, pT ≥ 1 GeV

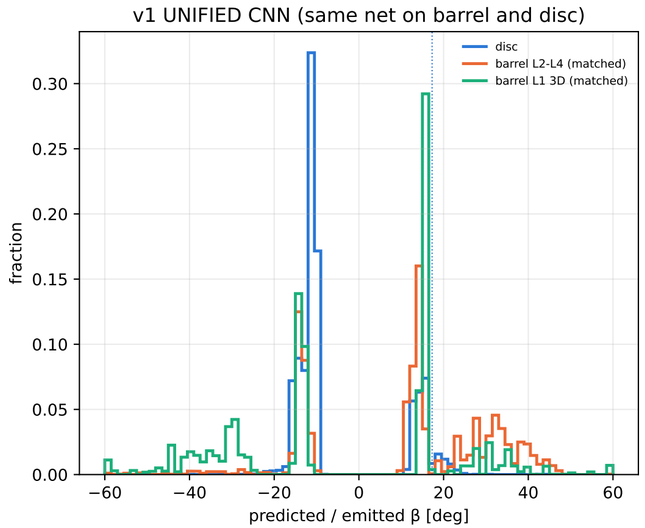


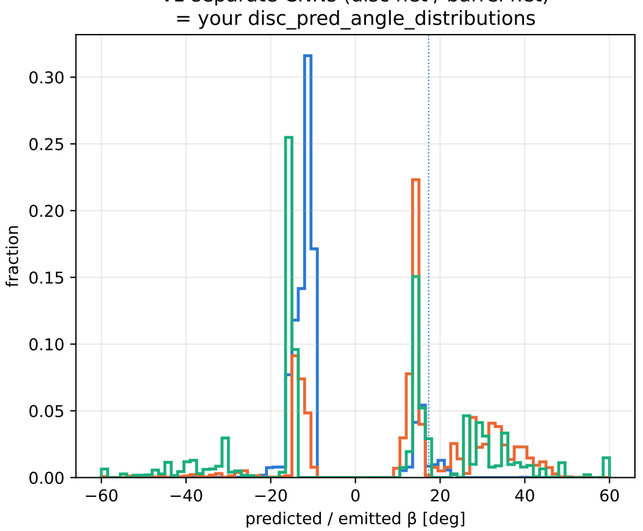


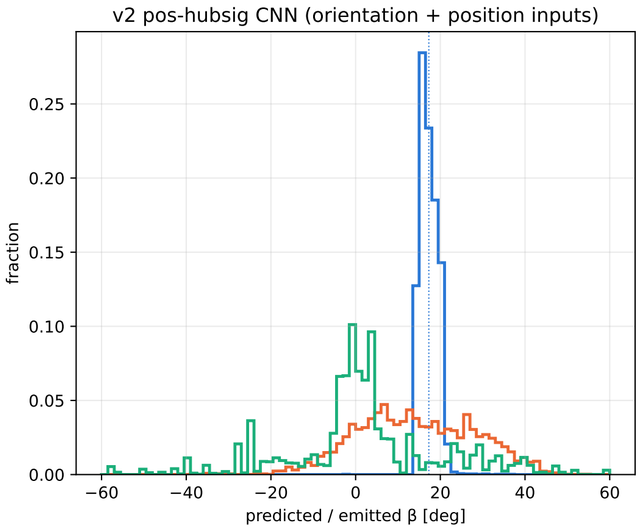


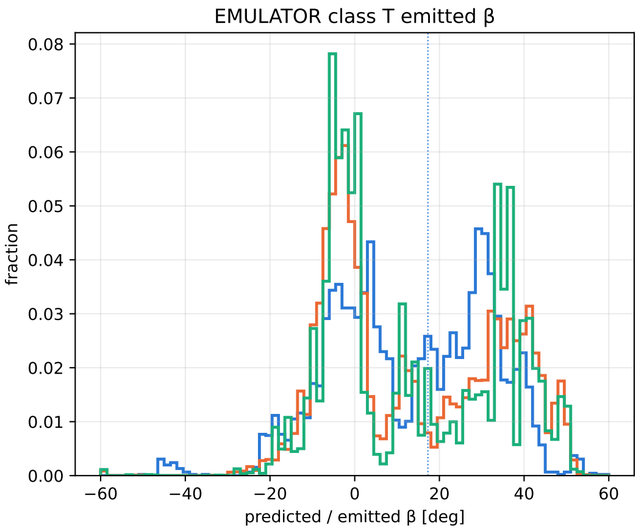


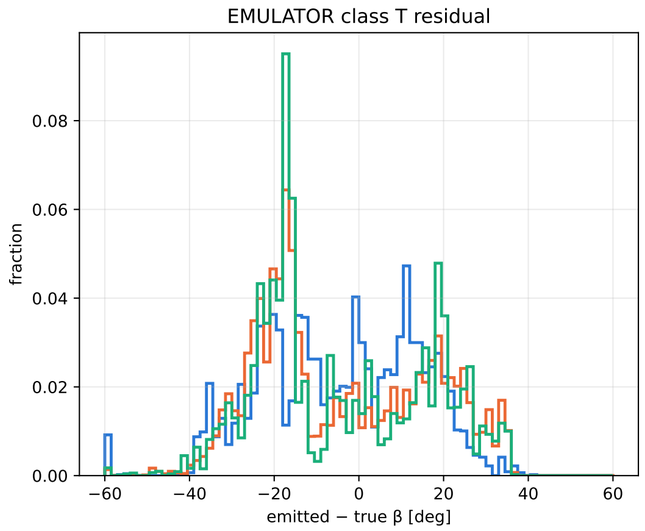


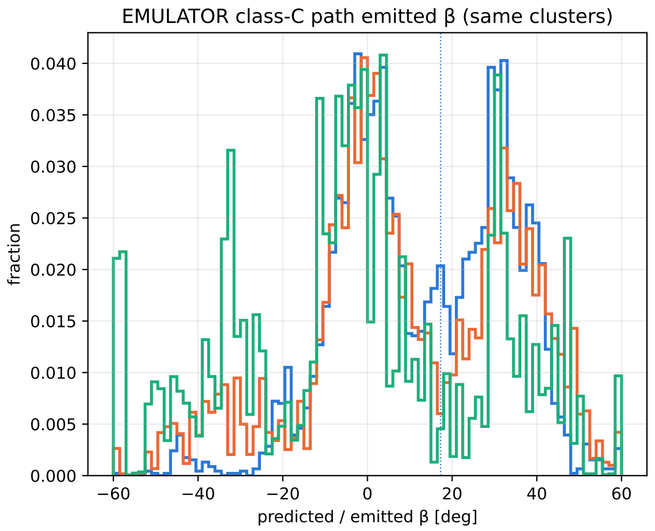








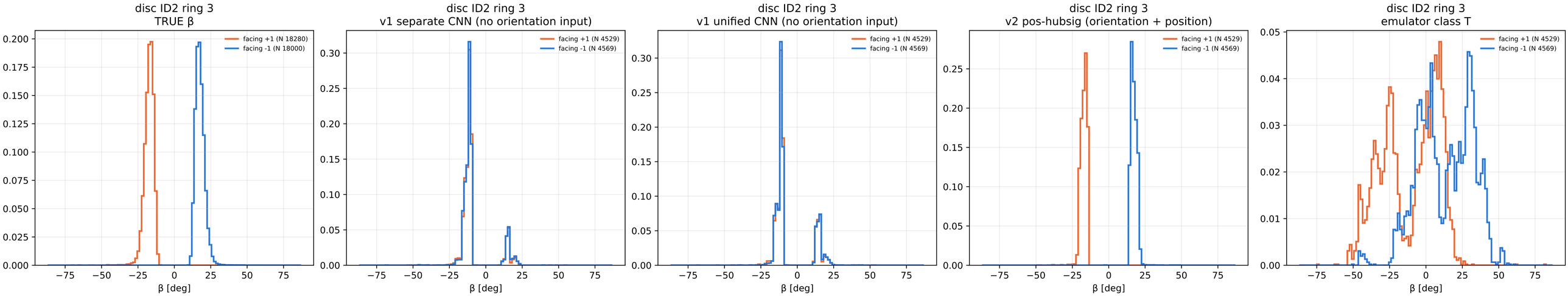


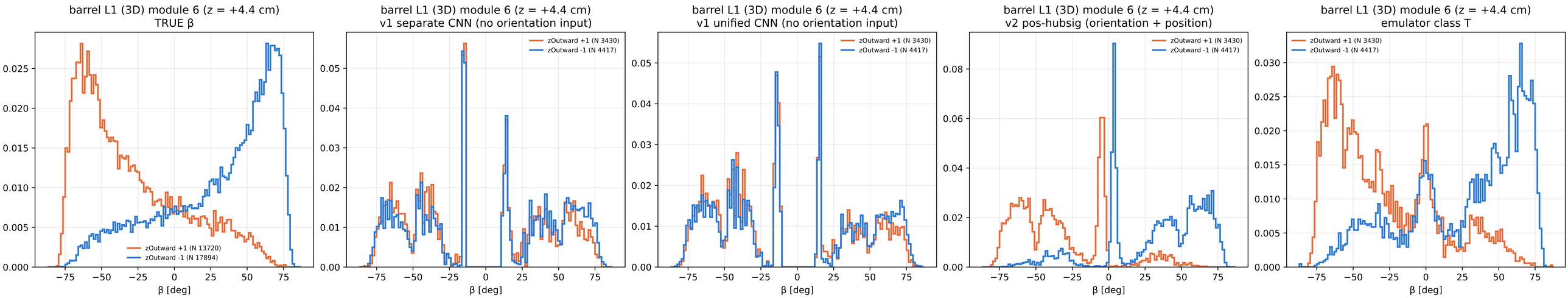


sample	N	Neff	P(sY≥2)	v1uni	v1sep	v2	emuT	core	opp
disc	18000	18000	0.397	-10.6	-10.8	17.0	14.4	18.0	0.28
L2-L4	321722	8797	0.403	14.3	14.4	13.5	9.2	20.5	0.39
L1	193151	1191	0.365	-12.9	13.1	0.8	9.9	20.2	0.39

best raw L2-L4 sensor: L4 module 4 zOutward +1 (W1 9.6 deg, N 5081)
best raw L1 sensor: L1 module 4 zOutward +1 (W1 30.9 deg, N 13426)
v1uni / v1sep / v2 / emuT = median predicted or emitted β [deg]
core = core RMS of emitted − true (mode-seeded 3σ clip, about 0) [deg]
opp = fraction emitted with the opposite sign to the truth
disc (β, α) weight in cells the barrel covers: L2-L4 1.000, L1 1.000
P(sY≥2): all 200 events; model columns: the 50 test events

Page 4. FLIPS. Same ring / same module ring, the two orientations: the truth is mirrored (sign(β) = −facing on the discs). A net that cannot see the orientation and gets no sign from the image (discs, L1: page 2) must give the SAME output for both: its sign is then right for one facing and wrong for the other. Model outputs: 50 test events.





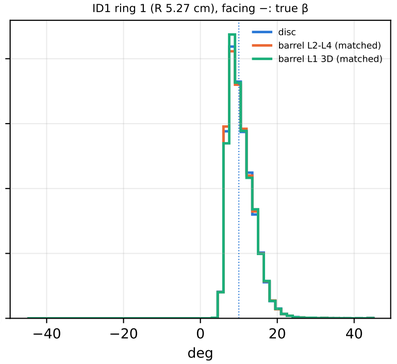
Sign accuracy and fraction predicted positive per surface × orientation (|true β| > 5 deg, test events)

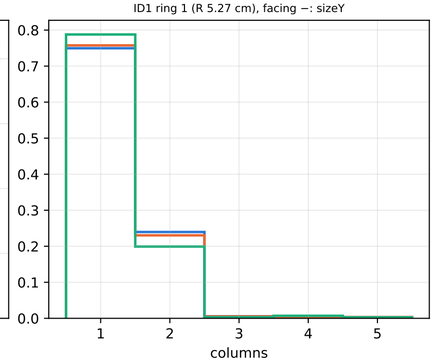
surface	facing	N	frac_true_pos	v1sep_frac_pos	v1sep_sign_acc	v1uni_frac_pos	v1uni_sign_acc	v2_frac_pos	v2_sign_acc	emuT_frac_pos	emuT_sign_acc
ID1	1	23231	0.000	0.366	0.634	0.198	0.802	0.000	1.000	0.405	0.595
ID1	-1	23044	1.000	0.375	0.375	0.210	0.210	1.000	1.000	0.778	0.778
ID2	1	23254	0.000	0.105	0.895	0.197	0.803	0.000	1.000	0.457	0.543
ID2	-1	23215	1.000	0.115	0.116	0.206	0.206	1.000	1.000	0.766	0.766
ID3	1	21156	0.000	0.245	0.754	0.167	0.833	0.000	1.000	0.487	0.513
ID3	-1	21453	1.000	0.259	0.259	0.183	0.183	1.000	1.000	0.772	0.772
ID4	1	18545	0.000	0.218	0.782	0.089	0.911	0.000	1.000	0.502	0.498
ID4	-1	18512	1.000	0.236	0.236	0.093	0.093	1.000	1.000	0.790	0.790
IL1	1	20427	0.505	0.538	0.607	0.480	0.605	0.501	0.909	0.500	0.939
IL1	-1	25934	0.496	0.540	0.500	0.466	0.503	0.496	0.907	0.490	0.941
IL2	1	15556	0.501	0.546	0.835	0.505	0.832	0.503	0.933	0.492	0.910
IL2	-1	17288	0.496	0.538	0.836	0.502	0.837	0.496	0.930	0.482	0.917
IL3	1	11278	0.501	0.518	0.801	0.537	0.798	0.499	0.936	0.486	0.878
IL3	-1	12407	0.497	0.503	0.799	0.531	0.794	0.496	0.932	0.484	0.883
IL4	1	9012	0.500	0.479	0.775	0.457	0.778	0.499	0.943	0.485	0.846
IL4	-1	9410	0.500	0.474	0.784	0.437	0.785	0.497	0.935	0.474	0.851

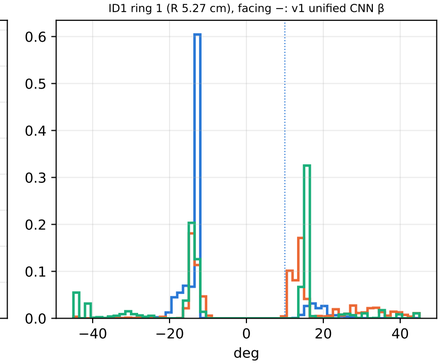
The emulator's disc offset. Class T emitted beta = truth + the response of the SmartPixels dump (Conv1D_Full-2bit, CMS axes) drawn at the lookup key. Discs are looked up at alpha_eq = atan(cotA - 0.312), where the PixelAV Lorentz drift cancels (the cluster has NO tilt, as on a real disc); the barrel at the true alpha (~0 near normal: the drift tilts the cluster). Inside the |beta| < ~20 deg plateau the dump leans POSITIVE at the disc key and is centred near 0 at the barrel key. On the discs sign(beta) = -facing, so facing - (beta > 0) gains and facing + (beta < 0) loses: inside the plateau the disc emitted medians sit +1.5 to +6 deg above the matched barrel ones (table pages). The sign carried into the CMS frame rests on the dump-to-CMS axis map (z_pix = -w_cms); CMSSW itself collects at w = -t/2 on every module. Dump cells hold 40-115 rows.

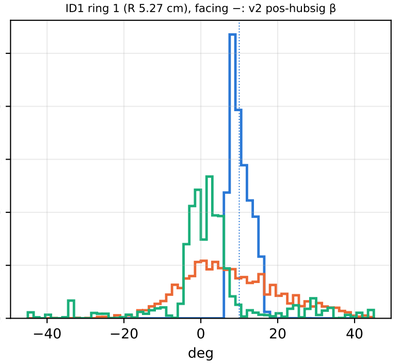
	alpha_band	true_beta	N	median_emitted_beta	median_residual	opposite_sign
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	-30..-20	110	-18.67	6.70	0.15
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	-20..-12	74	-3.32	13.39	0.39
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	-12..-6	56	-0.64	7.43	0.45
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	-6..0	51	2.57	4.70	0.71
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	0..6	57	4.46	2.73	0.05
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	6..12	74	4.91	-3.39	0.12
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	12..20	66	9.15	-5.44	0.15
disc key: alpha_eq = atan(cotA - 0.312), -21..-13 deg	(no tilt)	20..30	113	21.59	-3.53	0.06
barrel key near normal: alpha -4..4 deg		-30..-20	115	-23.01	3.03	0.10
barrel key near normal: alpha -4..4 deg		-20..-12	69	-7.26	9.56	0.22
barrel key near normal: alpha -4..4 deg		-12..-6	39	-4.24	4.15	0.26
barrel key near normal: alpha -4..4 deg		-6..0	38	0.08	2.51	0.53
barrel key near normal: alpha -4..4 deg		0..6	49	-0.25	-2.66	0.53
barrel key near normal: alpha -4..4 deg		6..12	61	0.92	-6.85	0.38
barrel key near normal: alpha -4..4 deg		12..20	72	2.29	-12.45	0.29
barrel key near normal: alpha -4..4 deg		20..30	104	20.16	-5.84	0.18

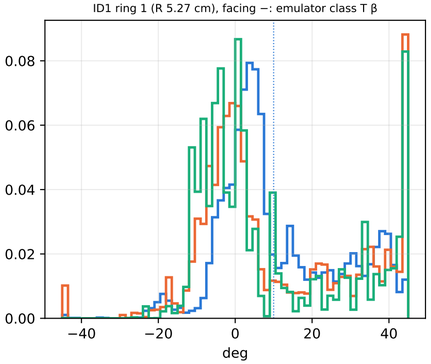
Pages 5-6. Two radii per disc (ring 1 = 5.3 cm, ring 4 = 13.8 cm), facing − (β > 0), vs barrel L2-L4 / L1 at matched incidence. pT ≥ 1 GeV. sizeY / true β: 200 events; model outputs: 50 test events.

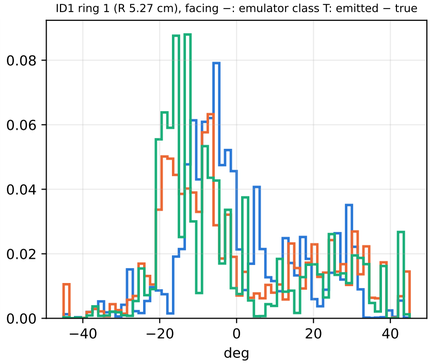


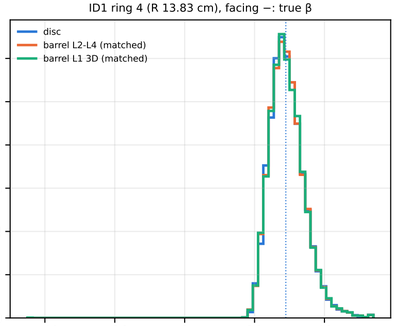


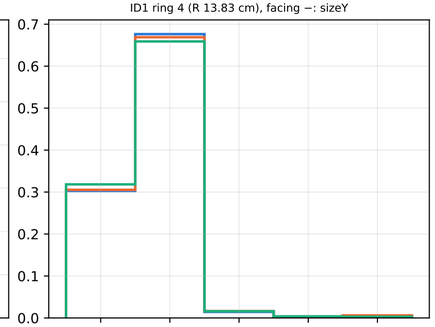


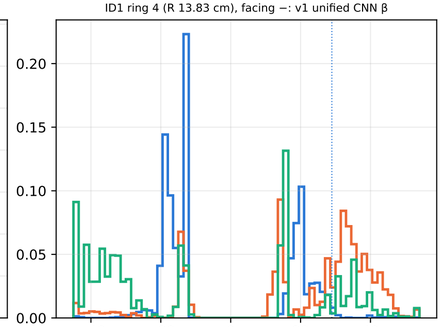


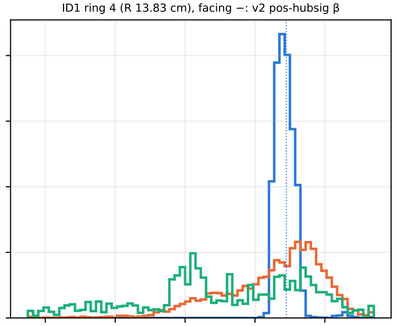


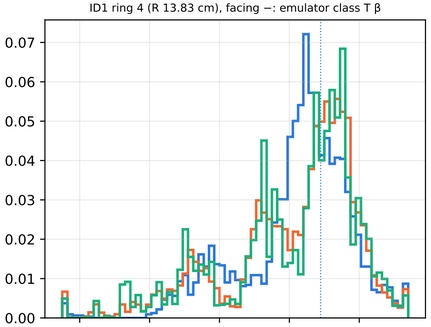


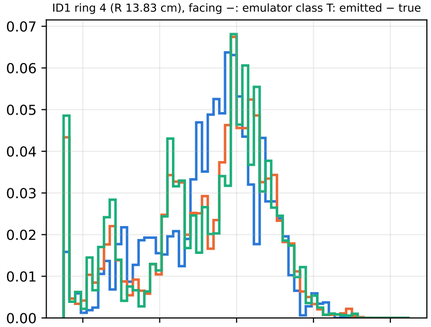


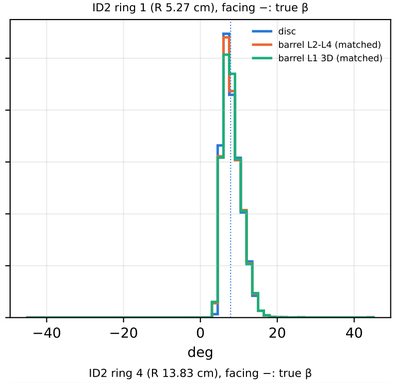


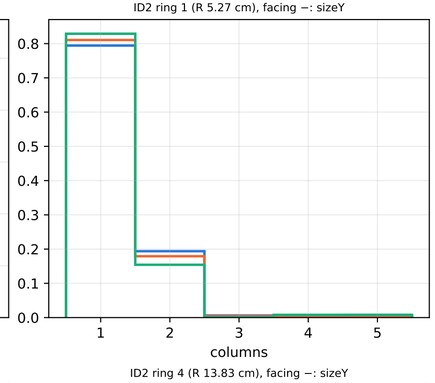


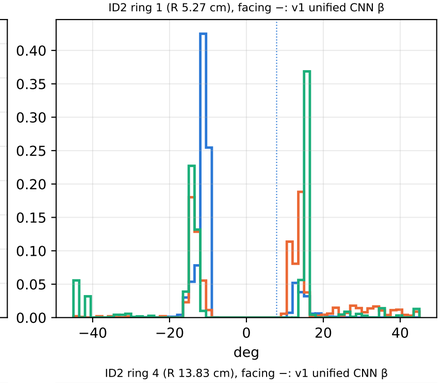


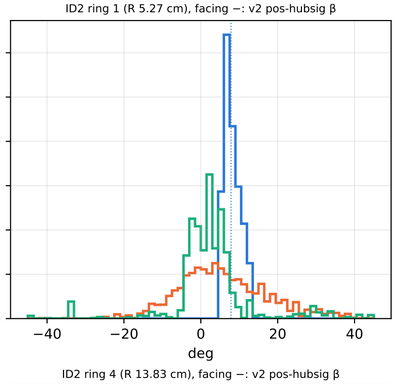


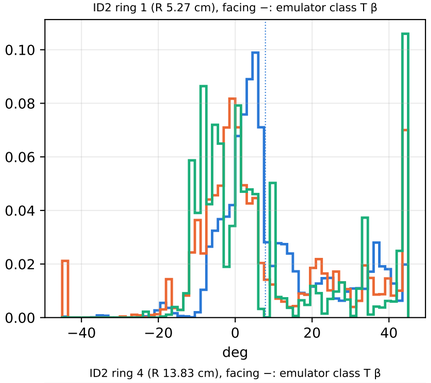


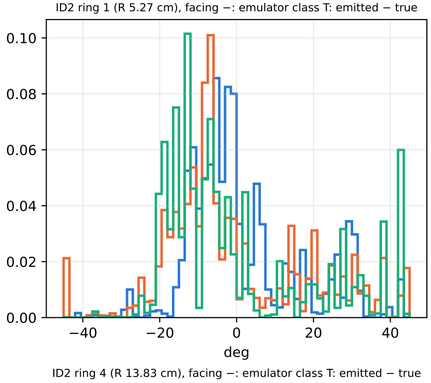


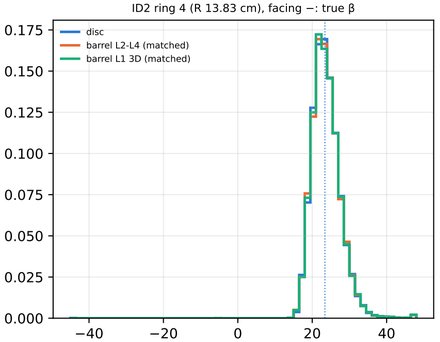


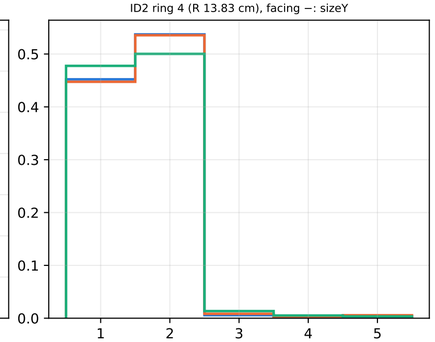


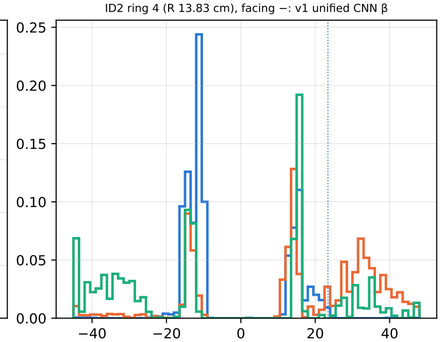


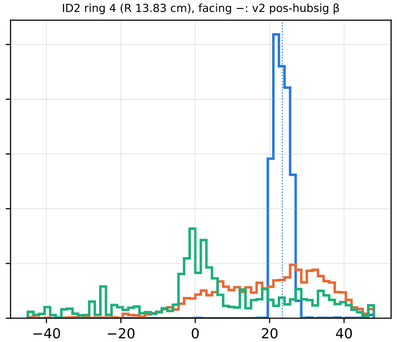


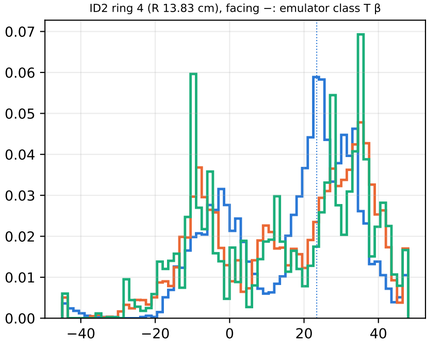


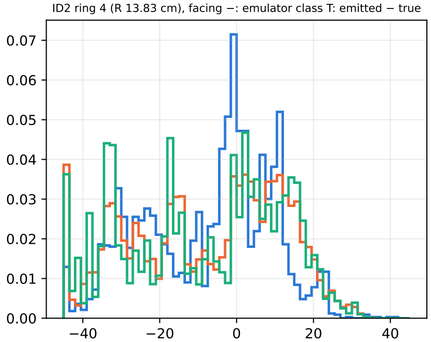




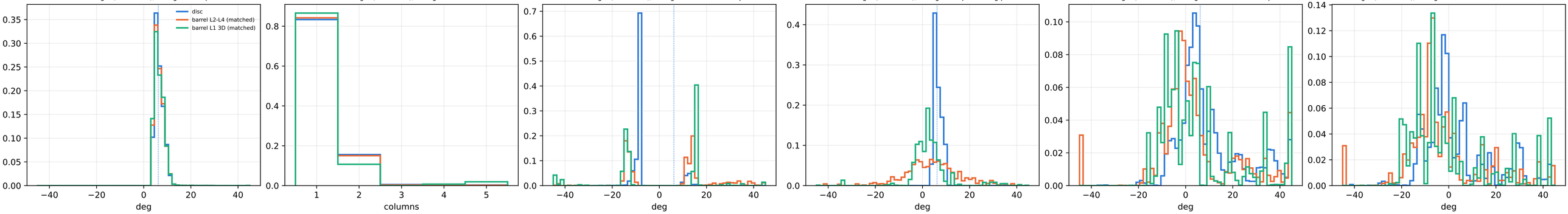


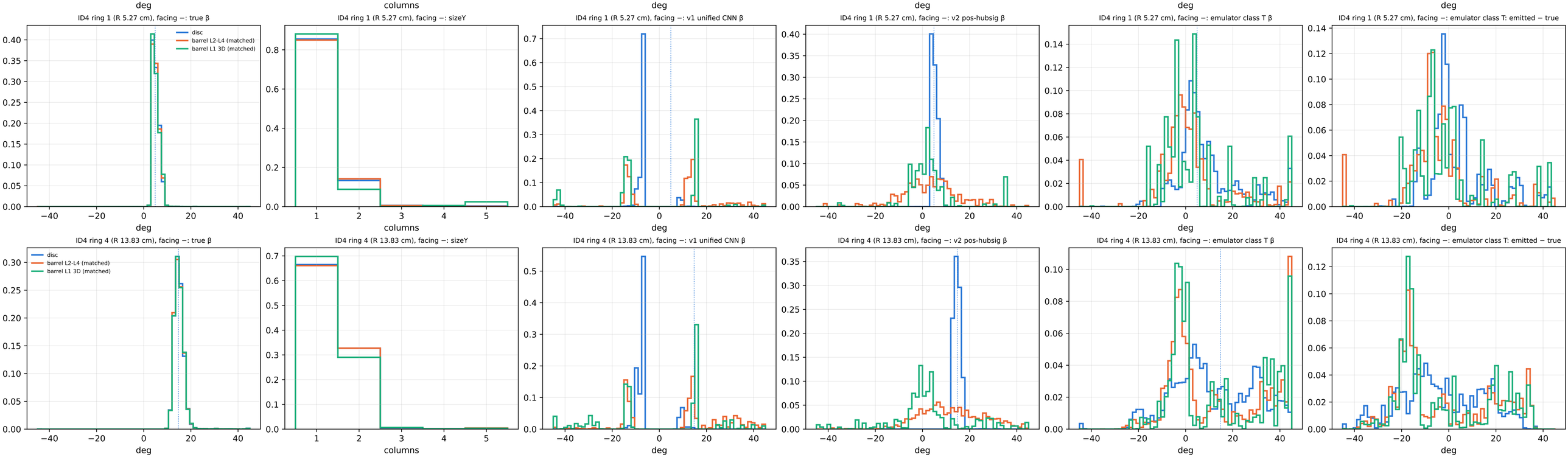






Pages 5-6. Two radii per disc (ring 1 = 5.3 cm, ring 4 = 13.8 cm), facing − (β > 0), vs barrel L2-L4 / L1 at matched incidence. pT ≥ 1 GeV. sizeY / true β: 200 events; model outputs: 50 test events.





Per disc sensor (disk, ring, facing) vs matched barrel: medians [deg], core / full RMS of emitted − true [deg], opposite-sign fractions

disk	ring	fac	sample	N	Neff	medβ	P(sY≥2)	yMin	v1uni	v1sep	v1sep_opp	v2	emuT	emuT_core	emuT_full	emuT_opp
2	3	-1	disc	18000	18000.000	17.283	0.397	0.277	-10.628	-10.786	0.852	17.019	14.369	18.011	18.902	0.280
2	3	-1	L2-L4	321722	8797.494	17.286	0.403	0.280	14.250	14.411	0.254	13.481	9.226	20.492	20.705	0.386
2	3	-1	L1 3D	193151	1190.819	17.298	0.365	0.271	-12.911	13.132	0.486	0.836	9.925	20.240	20.520	0.387
1	1	1	disc	34535	34535.000	-9.777	0.246	0.277	-12.924	-12.572	0.352	-9.542	0.184	20.109	20.109	0.504
1	1	1	L2-L4	321722	5035.221	-9.807	0.242	0.282	-11.303	10.598	0.517	-6.411	-4.695	17.263	17.310	0.328
1	1	1	L1 3D	193151	714.246	-9.635	0.234	0.289	14.921	-14.270	0.386	-1.303	-4.550	17.509	17.643	0.341
1	1	-1	disc	34450	34450.000	9.997	0.250	0.281	-12.924	-12.572	0.644	9.791	6.026	15.561	15.715	0.212
1	1	-1	L2-L4	321722	5102.558	9.996	0.242	0.278	12.213	13.884	0.324	6.633	2.295	18.930	21.273	0.407
1	1	-1	L1 3D	193151	730.541	9.973	0.212	0.277	-13.043	13.193	0.464	1.943	1.031	14.317	19.358	0.433
1	4	1	disc	12789	12789.000	-28.428	0.681	0.278	-13.274	-12.834	0.393	-27.950	-22.806	17.518	17.825	0.158
1	4	1	L2-L4	321722	21618.160	-28.423	0.677	0.278	-29.558	-29.572	0.258	-28.043	-29.014	16.172	18.117	0.132
1	4	1	L1 3D	193151	3129.464	-28.399	0.662	0.282	-14.806	-13.883	0.479	-7.611	-30.137	15.307	17.789	0.113
1	4	-1	disc	12508	12508.000	28.946	0.697	0.280	-13.274	-12.834	0.579	28.347	24.787	14.942	16.594	0.123
1	4	-1	L2-L4	321722	22468.243	28.935	0.695	0.276	30.188	30.212	0.153	28.419	27.063	16.858	19.590	0.157
1	4	-1	L1 3D	193151	3327.397	28.930	0.682	0.280	-15.177	-13.883	0.516	12.183	27.050	18.964	20.303	0.174
2	1	1	disc	34333	34333.000	-7.720	0.204	0.280	-10.628	-10.786	0.065	-7.593	1.038	18.933	18.933	0.525
2	1	1	L2-L4	321722	3020.774	-7.720	0.188	0.284	-11.078	11.159	0.541	-4.461	-4.103	15.874	15.878	0.304
2	1	1	L1 3D	193151	382.000	-7.667	0.176	0.278	14.921	-13.883	0.364	-1.413	-3.388	10.325	15.502	0.336
2	1	-1	disc	34439	34439.000	7.858	0.206	0.283	-10.628	-10.786	0.928	7.705	5.432	14.740	14.774	0.205
2	1	-1	L2-L4	321722	2874.220	7.889	0.189	0.275	11.464	13.004	0.345	4.664	1.320	16.911	20.777	0.434
2	1	-1	L1 3D	193151	427.329	7.871	0.171	0.285	-12.982	-13.883	0.504	2.335	1.032	12.533	19.669	0.436
2	4	1	disc	13582	13582.000	-22.999	0.536	0.277	-10.628	-10.786	0.192	-22.707	-19.510	20.192	20.260	0.298
2	4	1	L2-L4	321722	12019.250	-22.997	0.538	0.281	-19.170	-18.686	0.340	-21.531	-21.992	20.246	21.391	0.258
2	4	1	L1 3D	193151	1737.832	-22.993	0.522	0.283	-13.359	-13.883	0.495	-3.436	-24.052	19.064	21.814	0.242
2	4	-1	disc	13323	13323.000	23.411	0.548	0.277	-10.628	-10.786	0.786	23.074	21.734	17.550	18.671	0.240
2	4	-1	L2-L4	321722	12779.524	23.407	0.553	0.275	19.021	19.260	0.193	21.939	21.481	20.871	21.578	0.288
2	4	-1	L1 3D	193151	1857.727	23.360	0.522	0.275	-13.359	-13.883	0.513	3.349	20.124	21.856	22.681	0.301
3	1	1	disc	33802	33802.000	-6.077	0.170	0.280	-8.712	-8.689	0.218	-6.030	0.783	12.436	17.685	0.524
3	1	1	L2-L4	321722	1951.767	-6.106	0.142	0.285	-11.078	11.211	0.549	-3.114	-3.175	7.109	14.903	0.318
3	1	1	L1 3D	193151	238.239	-6.024	0.132	0.276	14.921	-13.883	0.341	-1.028	-0.121	7.999	12.728	0.437
3	1	-1	disc	33798	33798.000	6.173	0.166	0.281	-8.712	-8.689	0.771	6.087	5.318	13.369	14.380	0.178
3	1	-1	L2-L4	321722	1868.942	6.236	0.158	0.278	11.446	12.710	0.366	3.751	0.183	12.481	19.353	0.489
3	1	-1	L1 3D	193151	255.779	6.216	0.134	0.289	-11.757	-13.883	0.564	2.092	1.030	11.675	17.863	0.461
3	4	1	disc	13673	13673.000	-18.374	0.423	0.280	-8.712	-8.716	0.278	-18.159	-7.577	20.095	20.095	0.385
3	4	1	L2-L4	321722	6785.307	-18.370	0.427	0.279	-13.843	-12.798	0.416	-15.213	-12.164	20.857	21.509	0.340
3	4	1	L1 3D	193151	1068.929	-18.340	0.410	0.290	-13.359	14.031	0.561	-3.055	-11.069	21.204	23.306	0.354
3	4	-1	disc	13710	13710.000	18.605	0.431	0.280	-8.712	-8.689	0.689	18.315	16.397	18.399	19.616	0.294
3	4	-1	L2-L4	321722	7073.395	18.649	0.432	0.280	14.465	14.612	0.236	14.816	10.488	20.565	20.796	0.376
3	4	-1	L1 3D	193151	908.071	18.631	0.389	0.272	-13.359	13.274	0.479	1.117	10.685	20.480	20.825	0.356
4	1	1	disc	32872	32872.000	-4.795	0.143	0.282	-7.012	-6.996	0.203	-4.735	0.651	10.021	15.608	0.525
4	1	1	L2-L4	321722	1297.002	-4.800	0.113	0.285	-11.078	11.211	0.549	-1.486	-1.916	6.125	13.434	0.368
4	1	1	L1 3D	193151	179.153	-4.818	0.114	0.286	14.787	-13.883	0.363	1.285	0.399	5.891	10.638	0.547
4	1	-1	disc	32850	32850.000	4.867	0.145	0.281	-7.012	-6.996	0.799	4.858	5.377	11.157	13.719	0.162
4	1	-1	L2-L4	321722	1254.932	4.887	0.150	0.282	11.464	12.704	0.379	3.414	-0.649	9.653	18.120	0.527
4	1	-1	L1 3D	193151	161.232	4.737	0.119	0.288	-13.359	-13.883	0.592	1.826	1.961	10.318	15.684	0.467
4	4	1	disc	13918	13918.000	-14.588	0.325	0.281	-7.012	-6.996	0.250	-14.469	-1.951	21.141	21.141	0.453
4	4	1	L2-L4	321722	3794.319	-14.561	0.325	0.278	-11.283	-11.258	0.463	-9.050	-3.489	20.043	20.088	0.395
4	4	1	L1 3D	193151	567.237	-14.566	0.351	0.298	-14.294	-13.883	0.459	-1.837	-3.262	20.310	20.583	0.412
4	4	-1	disc	13599	13599.000	14.756	0.335	0.281	-7.012	-6.996	0.740	14.557	8.603	17.228	17.658	0.257
4	4	-1	L2-L4	321722	3986.478	14.746	0.339	0.286	13.648	14.253	0.283	11.172	2.590	20.504	20.518	0.416
4	4	-1	L1 3D	193151	516.062	14.855	0.302	0.262	14.805	13.132	0.489	0.900	1.028	16.745	20.067	0.420